Yousif Nazhat

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Education

Rutgers University — New Brunswick, NJ

Major: Information Technology & Informatics; Minor: Critical Intelligence — *Expected Graduation: May. 2027*

- **Relevant Coursework**: Data Structures, Computer Architecture, Discrete Structures, Linear Algebra

Professional Experience

RU Cybersecurity — New Brunswick, NJ

Team Lead — *Sep. 2025 – Present*

- Improved technical competency for 50+ active members as measured by a **30% increase in project participation**, by spearheading instructional workshops on **Red/Blue Team operations**, **SIEM**, and **Linux fundamentals**.
- Strengthened incident response skills as measured by the **successful execution of 4 simulated cyber-defense exercises**, by developing and leading infrastructure hardening and **threat mitigation workshops**.

RU Airborne, Avionics & Integration — New Brunswick, NJ

Design Team Lead — *Jan. 2025 – Present*

- Directed integration of interconnected avionics subsystems on a UAV **sponsored by NASA and Lockheed Martin**, applying rigorous interface validation to ensure **100% compliance** with strict safety and reliability standards.
- Authored comprehensive **technical reports** documenting testing methodology, data analysis, and system-level findings for stakeholder review.
- Delivered a fully integrated system within a strict **5-week timeline** by coordinating cross-functional technical teams across Aerodynamics and Manufacturing.
- Exceeded propulsion performance benchmarks by **12%**, surpassing the industry-standard thrust-to-weight target through **data-driven optimization** and iterative analysis.
- Maintained **zero critical system failures** during high-stress **130A peak-current operations** by developing and enforcing **standardized pre-operation checklists and safety protocols**.

Leadership & Involvement

Phi Mu Delta Fraternity — New Brunswick, NJ

Alumni Relations Chair — *Jan. 2024 – Present*

- Surpassed fundraising targets by **625%** for the Embrace Kids Foundation, securing sponsorships from **32 local organizations**.

Organizations

- **Phi Mu Delta Fraternity** - Alumni Chair — Sep. 2025 – Present
- **RU Airborne** - Avionics & Integration Design Team - Team Lead — Jan. 2025 – Present

Projects

Virtual Honeypot Security Monitoring Lab | *VirtualBox, Cowrie, ELK Stack*

- Simulated **credential-based intrusion attempts** by deploying a virtual honeypot environment that detected and logged **Nmap reconnaissance and Hydra brute-force activity**.
- Captured **1,000+ authentication attempts** and produced **actionable security analytics dashboards** to demonstrate threat-detection capabilities.

RISC-V ISA Subset Implementation | *C, Assembly*

- Achieved **100% execution fidelity** for the RISC-V ISA subset, measured by **passing all unit tests** for memory and arithmetic operations.
- Developed a **cycle-accurate 5-stage instruction pipeline** (Fetch, Decode, Execute, Memory, Writeback) and set-associative cache simulator.

Network Diagnostics & Security Analysis | *Python, Wireshark, Nmap*

- Analyzed DNS behavior and network traffic using **Wireshark and Nmap**, identifying vulnerabilities via **TCP/IP and OSI model analysis**.

Symbiote Host Compatibility Tree | *Java*

- Designed a **binary search tree** modeling compatibility operations for a biological data simulation.

Election Data Analysis Tool | *Java*

- Optimized **hierarchical data query efficiency** for thousands of records by architecting a **custom triply-nested linked list structure**.
- Parsed and aggregated CSV datasets **without external database libraries** for high-performance data retrieval.

128-bit Arithmetic Toolkit | *C*

- Enabled **high-precision cryptography-grade computation**, exceeding standard 64-bit hardware limits by **2x**.
- Engineered low-level bitwise algorithms (**shifting/masking**) to manipulate raw memory and custom structs.

Filesystem Emulator | *C*

- Created a **memory-based file system** supporting basic directory and file operations to simulate **OS-level storage management**.

Technical Skills

Languages & Tools: Java, C, Python, RISC-V Assembly, HTML/CSS, Wireshark, Nmap, GDB, Git, Linux/Unix, eCalc, Microsoft Office Suite

Cybersecurity: NIST CSF/RMF, SIEM (Splunk/ELK), Incident Response, Threat Modeling, Ransomware Prevention, Cyber Ops